

GENERIC REFINERY - GULF COAST

100 kBPSD Crude & Vacuum Distillation Unit

Design Dependency Map

Document No.	CFU-000-PM-DEP-001
Revision	A
Status	Issued for review (FEED)
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Prepared	Claude Code
Checked / Approved	- / -

1 Purpose

This document maps the engineering activities of the CDU/VDU design: the order they run in, what each one needs from upstream, and what it affects downstream. Use it to plan the remaining work and to judge the impact of a change. It covers the 45 activities that produce this package and the activities still needed before the design can be used for construction.

How to read it:

- An activity's **needs** are the activities whose results it uses. It can start when they have issued.
- A **wave** is a step in the sequence: wave 1 needs nothing, wave 2 needs only wave 1 results, and so on.

Activities in the same wave can run in parallel.

- An **iteration loop** is a later result that revises an earlier activity (for example, vendor data revising the mechanical calculations). Loops are why FEED takes more than one pass.

- **Downstream** means every activity reachable through the needs links. If an activity changes, all its downstream activities must be re-checked.

The network drawing is CFU-000-PM-DEP-002. The workbook has the full sequence, link list, design structure matrix (DSM), change-impact table, to-do list and code check. An interactive version is in the portal (dependencies.html).

2 Summary

Item	Value
Activities	45 (29 issued, 4 issued but provisional, 12 not started)
Sequence waves	13
Input links	129
Iteration loops	15
Deliverables mapped to activities	105
Data reads traced in the build	62 (62 upstream)
Code-check issues	0

3 Design sequence

Wave	ID	Activity	Discipline	Status	Needs
1	PRJ-01	Design basis (BOD)	Project	Issued (FEED)	-
2	PRJ-02	Crude assay characterisation	Process	Issued, provisional	PRJ-01
3	PRC-00	Rigorous simulation and model calibration	Process	Not started	PRJ-02
4	PRC-01	Heat and material balance	Process	Issued, provisional	PRJ-01, PRJ-02, PRC-00
5	PRC-02	Process equipment sizing	Process	Issued (FEED)	PRC-01
5	PRC-05	Block flow diagram	Process	Issued (FEED)	PRC-01
5	PIP-01	Piping material classes	Piping	Issued (FEED)	PRJ-01, PRC-01
6	PRC-03	Relief load and PSV sizing (unit)	Process	Issued (FEED)	PRC-01, PRC-02
6	PRC-04	Control loops and SIF definition	Process	Issued (FEED)	PRC-01, PRC-02
6	LAY-01	Plot plan and equipment layout	Layout	Issued (FEED)	PRC-02, PRC-01, PRJ-01

Wave	ID	Activity	Discipline	Status	Needs
6	ELE-01	Electrical load list	Electrical	Issued (FEED)	PRC-02
7	PRC-06	Process flow diagrams	Process	Issued (FEED)	PRC-01, PRC-02, PRC-04
7	PRC-07	Process design report	Process	Issued (FEED)	PRC-01, PRC-02, PRC-03
7	SAF-02	Global flare load study	Safety	Not started	PRC-03
7	MEC-01	Mechanical design calculations	Mechanical	Issued (FEED)	PRC-02, PRC-01, PRC-03
7	LAY-02	Sections and 3D layout model	Layout	Issued (FEED)	LAY-01, PRC-02
7	ICS-01	Control philosophy and schemes	I&C	Issued (FEED)	PRC-04, PRC-01, PRC-02
7	ELE-02	Hazardous area classification	Electrical	Issued (FEED)	LAY-01, PRC-02, PRC-04
7	ELE-03	Electrical sizing, single-line diagrams	Electrical	Issued (FEED)	ELE-01
8	PRC-08	P&IDs, line list, instrument index	Process	Issued (FEED)	PRC-02, PRC-03, PRC-04, PRC-06, PIP-01, LAY-01
8	MEC-02	Equipment datasheets	Mechanical	Issued (FEED)	MEC-01, PRC-02, PRC-03
8	MEC-03	Equipment GA drawings	Mechanical	Issued (FEED)	MEC-01
8	ELE-04	Cable schedule	Electrical	Issued (FEED)	ELE-01, ELE-03, LAY-01, ELE-02
8	CST-01	Class 4 cost estimate	Cost	Issued (FEED)	PRC-02, MEC-01
9	SAF-01	Preliminary HAZOP	Safety	Issued (FEED)	PRC-08, PRC-04, PRC-03
9	VEN-01	Vendor enquiries and vendor data	Mechanical	Not started	MEC-02
9	PIP-02	Piping routing and 3D model	Piping	Issued (FEED)	PRC-08, LAY-01, PRC-03, MEC-01
9	ICS-04	Control valve sizing	I&C	Issued (FEED)	PRC-01, PRC-02, PRC-04, PRC-08
9	ICS-05	I/O list	I&C	Issued (FEED)	PRC-08, PRC-04, PRC-02, LAY-01
10	SAF-04	Siting study, QRA, F&G mapping	Safety	Not started	LAY-01, SAF-01, ELE-02
10	PIP-03	Pipe stress and supports (screening)	Piping	Issued, provisional	PIP-02, MEC-01
10	PIP-04	Isometrics and piping MTO	Piping	Issued (FEED)	PIP-02, PIP-01
10	ICS-02	SIF list and SIL determination (LOPA)	I&C	Issued, provisional	PRC-04, SAF-01
10	ICS-06	Loop diagrams	I&C	Issued (FEED)	ICS-05
10	ICS-07	ICS architecture	I&C	Issued (FEED)	ICS-05, ELE-03
10	ELE-05	Power system studies and protection	Electrical	Not started	ELE-03, VEN-01
11	SAF-03	Formal HAZOP and LOPA workshop	Safety	Not started	PRC-08, SAF-01, ICS-02
11	PIP-05	Detailed pipe stress (CAESAR II)	Piping	Not started	PIP-03, VEN-01
11	CIV-01	Geotechnical, civil and structural design	Civil	Not started	LAY-01, MEC-01, PIP-03

Wave	ID	Activity	Discipline	Status	Needs
11	ICS-03	Cause and effect matrix	I&C	Issued (FEED)	PRC-04, PRC-03, PRC-02, ICS-02
12	PRJ-03	Document register and portal	Project	Issued (FEED)	CST-01, SAF-01, PIP-04, ICS-06, ICS-07, ELE-04, MEC-03, LAY-02, PRC-05, PRC-07, ICS-01, ICS-03, ICS-04, MEC-02, PIP-03
12	LAY-03	3D model reviews and constructability	Layout	Not started	PIP-02, LAY-02, CIV-01
12	ICS-08	SIL verification and SRS	I&C	Not started	ICS-02, SAF-03, VEN-01
12	CST-02	Class 3 cost estimate	Cost	Not started	CST-01, VEN-01, PIP-04, ELE-04, CIV-01
13	PRJ-04	PE review, seal and IFC issue	Project	Not started	PRJ-03, SAF-03, ICS-08, PIP-05, ELE-05, CST-02, LAY-03, SAF-04, SAF-02

4 Inputs and impacts by activity

For each activity: what it needs (and what flows across each link), what it needs from outside the design team, and what it feeds. Downstream counts are the activities and deliverables to re-check if it changes.

PRJ-01 Design basis (BOD) (Project, issued (feed))

- External inputs: Owner capacity, crude slate, product specs, turndown; Site, climate and utility conditions; Codes, standards and owner specifications
- Issues: CFU-000-PR-BOD-001
- Feeds: PRJ-02, PRC-01, PIP-01, LAY-01; 44 downstream activities, 104 downstream deliverables
- Note: Every other activity reads the basis; a change here re-opens the whole package.

PRJ-02 Crude assay characterisation (Process, issued, provisional)

- Needs: PRJ-01 (crude slate, design and check crudes)
- External inputs: Full laboratory assay (TBP, gravity, sulphur, salts, metals, acidity)
- Feeds: PRC-00, PRC-01; 43 downstream activities, 104 downstream deliverables
- Note: Uses a published Arab Light assay with pseudo-components, not a lab assay.

PRC-00 Rigorous simulation and model calibration (Process, not started)

- Needs: PRJ-02 (lab assay pseudo-components)
- External inputs: Simulator licence (HYSYS / Petro-SIM / Pro-II); Licensor or owner operating data
- Feeds: PRC-01; 42 downstream activities, 104 downstream deliverables
- Note: Replaces the ideal-VLE shortcut model; expect 5-10 C changes in draw temperatures.

PRC-01 Heat and material balance (Process, issued, provisional)

- Needs: PRJ-01 (capacity, product specs, utility conditions); PRJ-02 (pseudo-components); PRC-00 (calibrated cut points and efficiencies)
- Owns data: data/streams.json, data/process_results.json
- Issues: CFU-000-PR-HMB-001

- Feeds: PRC-02, PRC-05, PIP-01, PRC-03, PRC-04, LAY-01, PRC-06, PRC-07, MEC-01, ICS-01, ICS-04; 41 downstream activities, 103 downstream deliverables

PRC-02 Process equipment sizing (Process, issued (feed))

- Needs: PRC-01 (flows, duties, column vapour/liquid loads)
- Owns data: data/equipment.json
- Issues: CFU-000-ME-LST-001
- Feeds: PRC-03, PRC-04, LAY-01, ELE-01, PRC-06, PRC-07, MEC-01, LAY-02, ICS-01, ELE-02, PRC-08, MEC-02, CST-01, ICS-04, ICS-05, ICS-03; 38 downstream activities, 100 downstream deliverables
- Revised by: MEC-01 (mechanical holds re-rate sizing (C-201 bed-1 diameter, heater box length)); PIP-02 (routed lengths update pump and line hydraulics)
- Note: Equipment list is the hub: mechanical, layout, electrical, I&C and cost all read it.

PRC-05 Block flow diagram (Process, issued (feed))

- Needs: PRC-01 (unit yields)
- Issues: CFU-000-PR-BFD-001
- Feeds: PRJ-03; 2 downstream activities, 4 downstream deliverables

PIP-01 Piping material classes (Piping, issued (feed))

- Needs: PRJ-01 (codes, corrosion basis); PRC-01 (design T/P, sulphur and acid corrosion)
- Issues: CFU-000-PI-SPC-001
- Feeds: PRC-08, PIP-04; 20 downstream activities, 51 downstream deliverables
- Note: Defined in CONVENTIONS.md; summary issued with the P&IDs.

PRC-03 Relief load and PSV sizing (unit) (Process, issued (feed))

- Needs: PRC-01 (relief compositions and rates); PRC-02 (design pressures, wetted areas)
- Owns data: data/psv.json
- Feeds: PRC-07, SAF-02, MEC-01, PRC-08, MEC-02, SAF-01, PIP-02, ICS-03; 28 downstream activities, 70 downstream deliverables
- Revised by: SAF-02 (flare back-pressure changes PSV type and size)

PRC-04 Control loops and SIF definition (Process, issued (feed))

- Needs: PRC-01 (operating points); PRC-02 (equipment and instrument ranges)
- Owns data: data/control_loops.json
- Feeds: PRC-06, ICS-01, ELE-02, PRC-08, SAF-01, ICS-04, ICS-05, ICS-02, ICS-03; 24 downstream activities, 70 downstream deliverables

LAY-01 Plot plan and equipment layout (Layout, issued (feed))

- Needs: PRC-02 (equipment list and footprints); PRC-01 (hazard inventory, heater duties); PRJ-01 (site, spacing standards)
- Owns data: data/layout.json
- Issues: CFU-000-PL-PLT-001
- Feeds: LAY-02, ELE-02, PRC-08, ELE-04, PIP-02, ICS-05, SAF-04, CIV-01; 23 downstream activities, 59 downstream deliverables
- Revised by: MEC-01 (calculated dimensions and weights update footprints and spacing); PIP-02 (rack width and clashes move equipment); SAF-04 (siting study moves occupied buildings and equipment); VEN-01 (vendor package footprints)

ELE-01 Electrical load list (Electrical, issued (feed))

- Needs: PRC-02 (motor ratings, absorbed power)
- Issues: CFU-000-EL-LDL-001
- Feeds: ELE-03, ELE-04; 7 downstream activities, 11 downstream deliverables
- Revised by: ICS-07 (ICS cabinet and UPS loads added to the load list); VEN-01 (certified motor ratings)

PRC-06 Process flow diagrams (Process, issued (feed))

- Needs: PRC-01 (stream table); PRC-02 (equipment data); PRC-04 (principal control loops)
- Issues: CFU-000-PR-PFD-ALL, CFU-100-PR-PFD-001, CFU-100-PR-PFD-002, CFU-100-PR-PFD-003, CFU-100-PR-PFD-004, CFU-200-PR-PFD-005, CFU-200-PR-PFD-006
- Feeds: PRC-08; 20 downstream activities, 51 downstream deliverables

PRC-07 Process design report (Process, issued (feed))

- Needs: PRC-01 (H&MB); PRC-02 (sizing results); PRC-03 (relief summary)
- Issues: CFU-000-PR-RPT-001
- Feeds: PRJ-03; 2 downstream activities, 4 downstream deliverables

SAF-02 Global flare load study (Safety, not started)

- Needs: PRC-03 (unit relief loads)
- External inputs: Refinery flare network model, other units' loads
- Feeds: PRJ-04; 1 downstream activities, 0 downstream deliverables
- Note: Governing case (C-101 reflux failure, about 220 t/h) needs power and cooling-water failure cases.

MEC-01 Mechanical design calculations (Mechanical, issued (feed))

- Needs: PRC-02 (design P/T, dimensions, internals); PRC-01 (nozzle flows); PRC-03 (PSV set points)
- Owns data: data/mech.json
- Issues: CFU-000-ME-CAL-001
- Feeds: MEC-02, MEC-03, CST-01, PIP-02, PIP-03, CIV-01; 15 downstream activities, 38 downstream deliverables
- Revised by: VEN-01 (certified vendor dimensions, weights and nozzles)

LAY-02 Sections and 3D layout model (Layout, issued (feed))

- Needs: LAY-01 (equipment positions, structures); PRC-02 (equipment heights)
- Issues: CFU-000-PL-3DM-001, CFU-000-PL-ELV-001, CFU-000-PL-ELV-002, CFU-000-PL-ELV-003
- Feeds: PRJ-03, LAY-03; 3 downstream activities, 4 downstream deliverables

ICS-01 Control philosophy and schemes (I&C, issued (feed))

- Needs: PRC-04 (loops and SIFs); PRC-01 (operating points); PRC-02 (equipment)
- Issues: CFU-000-IC-CSD-006, CFU-000-IC-CSD-ALL, CFU-000-IC-RPT-001, CFU-100-IC-CSD-001, CFU-100-IC-CSD-002, CFU-100-IC-CSD-003, CFU-100-IC-CSD-004, CFU-200-IC-CSD-005
- Feeds: PRJ-03; 2 downstream activities, 4 downstream deliverables

ELE-02 Hazardous area classification (Electrical, issued (feed))

- Needs: LAY-01 (release-source positions); PRC-02 (fluids and conditions); PRC-04 (analysers and vents)
- Issues: CFU-000-EL-HAC-001, CFU-000-EL-HAC-002, CFU-000-EL-HAC-003
- Feeds: ELE-04, SAF-04; 5 downstream activities, 5 downstream deliverables

ELE-03 Electrical sizing, single-line diagrams (Electrical, issued (feed))

- Needs: ELE-01 (loads and demand)

- Owns data: data/electrical.json
- Issues: CFU-000-EL-CAL-001, CFU-000-EL-SLD-001, CFU-000-EL-SLD-002, CFU-000-EL-SLD-003, CFU-000-EL-SLD-004
- Feeds: ELE-04, ICS-07, ELE-05; 6 downstream activities, 6 downstream deliverables
- Revised by: ELE-04 (actual cable impedances re-run the motor-starting voltage-dip check)

PRC-08 P&IDs, line list, instrument index (Process, issued (feed))

- Needs: PRC-02 (equipment and design conditions); PRC-03 (PSV tags and sizes); PRC-04 (control loops and SIFs); PRC-06 (PFD topology); PIP-01 (piping classes); LAY-01 (fire and gas detector locations)
- Owns data: data/lines.json, data/instruments.json
- Issues: CFU-000-IC-IDX-001, CFU-000-PI-LL-001, CFU-000-PR-PID-000, CFU-000-PR-PID-ALL, CFU-100-PR-PID-001, CFU-100-PR-PID-002, CFU-100-PR-PID-003, CFU-100-PR-PID-004, CFU-100-PR-PID-005, CFU-100-PR-PID-006, CFU-100-PR-PID-007, CFU-100-PR-PID-008, CFU-100-PR-PID-009, CFU-100-PR-PID-010, CFU-100-PR-PID-011, CFU-200-PR-PID-012, CFU-200-PR-PID-013, CFU-200-PR-PID-014, CFU-200-PR-PID-015, CFU-900-PR-PID-016
- Feeds: SAF-01, PIP-02, ICS-04, ICS-05, SAF-03; 19 downstream activities, 31 downstream deliverables
- Revised by: ICS-04 (control valve sizes and reducers shown on the P&IDs); SAF-01 (HAZOP recommendations revise the P&IDs); SAF-03 (formal HAZOP actions revise the P&IDs)
- Note: Line list and instrument index feed piping, I&C and electrical; P&ID revisions ripple widely.

MEC-02 Equipment datasheets (Mechanical, issued (feed))

- Needs: MEC-01 (thicknesses, weights, nozzles); PRC-02 (process data); PRC-03 (PSV data)
- Issues: CFU-000-ME-DS-000, CFU-000-ME-DS-001, CFU-000-ME-DS-002, CFU-000-ME-DS-003, CFU-000-ME-DS-004, CFU-000-ME-DS-005, CFU-000-ME-DS-006, CFU-000-ME-DS-007
- Feeds: VEN-01, PRJ-03; 7 downstream activities, 4 downstream deliverables

MEC-03 Equipment GA drawings (Mechanical, issued (feed))

- Needs: MEC-01 (geometry, nozzle schedule)
- Issues: CFU-100-ME-GA-001, CFU-100-ME-GA-002, CFU-100-ME-GA-003, CFU-100-ME-GA-005, CFU-100-ME-GA-007, CFU-100-ME-GA-008, CFU-200-ME-GA-004, CFU-200-ME-GA-006
- Feeds: PRJ-03; 2 downstream activities, 4 downstream deliverables

ELE-04 Cable schedule (Electrical, issued (feed))

- Needs: ELE-01 (loads); ELE-03 (feeders); LAY-01 (routes); ELE-02 (zone ratings)
- Issues: CFU-000-EL-CBL-001
- Feeds: PRJ-03, CST-02; 3 downstream activities, 4 downstream deliverables

CST-01 Class 4 cost estimate (Cost, issued (feed))

- Needs: PRC-02 (equipment list); MEC-01 (shell weights)
- Issues: CFU-000-PM-EST-001
- Feeds: PRJ-03, CST-02; 3 downstream activities, 4 downstream deliverables

SAF-01 Preliminary HAZOP (Safety, issued (feed))

- Needs: PRC-08 (P&IDs); PRC-04 (SIFs); PRC-03 (relief devices)
- Issues: CFU-000-PR-RPT-002
- Feeds: SAF-04, ICS-02, SAF-03, PRJ-03; 7 downstream activities, 6 downstream deliverables

VEN-01 Vendor enquiries and vendor data (Mechanical, not started)

- Needs: MEC-02 (datasheets for enquiry)

- External inputs: Vendor quotations and certified data (heaters, desalters, packing, pumps, ejectors, air coolers)
- Feeds: ELE-05, PIP-05, ICS-08, CST-02; 5 downstream activities, 0 downstream deliverables
- Note: Vendor data revises weights, nozzles, motor ratings and I/O; plan for one full update cycle.

PIP-02 Piping routing and 3D model (Piping, issued (feed))

- Needs: PRC-08 (line list, in-line instruments); LAY-01 (nozzle positions, pipe rack); PRC-03 (PSV inlet/outlet); MEC-01 (nozzle sizes, weights)
- Owns data: data/routing.json
- Issues: CFU-000-PI-3DM-002
- Feeds: PIP-03, PIP-04, LAY-03; 8 downstream activities, 20 downstream deliverables

ICS-04 Control valve sizing (I&C, issued (feed))

- Needs: PRC-01 (flows, densities); PRC-02 (pump curves); PRC-04 (valve tags); PRC-08 (line sizes)
- Issues: CFU-000-IC-CAL-001
- Feeds: PRJ-03; 2 downstream activities, 4 downstream deliverables

ICS-05 I/O list (I&C, issued (feed))

- Needs: PRC-08 (instrument index); PRC-04 (loops and SIFs); PRC-02 (motor drivers); LAY-01 (junction-box zones)
- Owns data: data/io_list.json
- Issues: CFU-000-IC-IOL-001
- Feeds: ICS-06, ICS-07; 4 downstream activities, 9 downstream deliverables
- Revised by: VEN-01 (vendor package I/O)

SAF-04 Siting study, QRA, F&G mapping (Safety, not started)

- Needs: LAY-01 (plot plan); SAF-01 (major hazards); ELE-02 (release sources)
- External inputs: Occupied building list, meteorology
- Feeds: PRJ-04; 1 downstream activities, 0 downstream deliverables
- Note: API 752/753 siting, QRA, fire and gas mapping and fireproofing; may move buildings or equipment.

PIP-03 Pipe stress and supports (screening) (Piping, issued, provisional)

- Needs: PIP-02 (routed geometry); MEC-01 (nozzle allowables)
- Issues: CFU-000-PI-RPT-001
- Feeds: PIP-05, CIV-01, PRJ-03; 6 downstream activities, 4 downstream deliverables
- Note: Screening only; CAESAR II analysis is PIP-05.

PIP-04 Isometrics and piping MTO (Piping, issued (feed))

- Needs: PIP-02 (routed geometry); PIP-01 (piping classes)
- Issues: CFU-000-PI-ISO-000, CFU-000-PI-MTO-001, CFU-100-PI-ISO-001, CFU-100-PI-ISO-002, CFU-100-PI-ISO-003, CFU-100-PI-ISO-004, CFU-100-PI-ISO-005, CFU-100-PI-ISO-006, CFU-100-PI-ISO-007, CFU-100-PI-ISO-008, CFU-100-PI-ISO-009, CFU-100-PI-ISO-010, CFU-100-PI-ISO-011, CFU-100-PI-ISO-012, CFU-200-PI-ISO-001
- Feeds: PRJ-03, CST-02; 3 downstream activities, 4 downstream deliverables

ICS-02 SIF list and SIL determination (LOPA) (I&C, issued, provisional)

- Needs: PRC-04 (SIF definitions); SAF-01 (hazard scenarios)
- Issues: CFU-000-IC-RPT-002
- Feeds: SAF-03, ICS-03, ICS-08; 5 downstream activities, 5 downstream deliverables

- Note: LOPA frequencies are FEED judgements pending SAF-03.

ICS-06 Loop diagrams (I&C, issued (feed))

- Needs: ICS-05 (terminations, cabinets)
- Issues: CFU-100-IC-LD-001, CFU-100-IC-LD-002, CFU-100-IC-LD-003, CFU-100-IC-LD-ALL
- Feeds: PRJ-03; 2 downstream activities, 4 downstream deliverables

ICS-07 ICS architecture (I&C, issued (feed))

- Needs: ICS-05 (I/O counts); ELE-03 (UPS feeds)
- Issues: CFU-000-IC-BLK-001
- Feeds: PRJ-03; 2 downstream activities, 4 downstream deliverables

ELE-05 Power system studies and protection (Electrical, not started)

- Needs: ELE-03 (SLD); VEN-01 (motor and transformer data)
- External inputs: Utility short-circuit data; ETAP or equivalent
- Feeds: PRJ-04; 1 downstream activities, 0 downstream deliverables
- Note: Load flow, short circuit, motor starting, arc flash, relay coordination.

SAF-03 Formal HAZOP and LOPA workshop (Safety, not started)

- Needs: PRC-08 (IFD P&IDs); SAF-01 (preliminary findings); ICS-02 (draft SIL targets)
- External inputs: Multidisciplinary team incl. owner operations
- Feeds: ICS-08, PRJ-04; 2 downstream activities, 0 downstream deliverables

PIP-05 Detailed pipe stress (CAESAR II) (Piping, not started)

- Needs: PIP-03 (critical line list); VEN-01 (certified nozzle allowables)
- External inputs: CAESAR II or equivalent
- Feeds: PRJ-04; 1 downstream activities, 0 downstream deliverables
- Note: Includes nozzle-load agreement with equipment vendors.

CIV-01 Geotechnical, civil and structural design (Civil, not started)

- Needs: LAY-01 (plot plan); MEC-01 (operating and test weights); PIP-03 (support loads)
- External inputs: Geotechnical survey; Topographic survey
- Feeds: LAY-03, CST-02; 3 downstream activities, 0 downstream deliverables
- Note: No civil deliverables in this package: foundations, pipe rack and structures are still to do.

ICS-03 Cause and effect matrix (I&C, issued (feed))

- Needs: PRC-04 (SIFs); PRC-03 (relief devices); PRC-02 (equipment); ICS-02 (SIL targets)
- Issues: CFU-000-IC-CE-001
- Feeds: PRJ-03; 2 downstream activities, 4 downstream deliverables

PRJ-03 Document register and portal (Project, issued (feed))

- Needs: CST-01 (estimate); SAF-01 (HAZOP); PIP-04 (isometrics); ICS-06 (loop diagrams); ICS-07 (architecture); ELE-04 (cables); MEC-03 (GAs); LAY-02 (3D model); PRC-05 (BFD); PRC-07 (process report); ICS-01 (control philosophy); ICS-03 (C&E); ICS-04 (CV sizing); MEC-02 (datasheets); PIP-03 (stress study)
- Issues: CFU-000-PM-DEP-001, CFU-000-PM-DEP-002, CFU-000-PM-DEP-003, CFU-000-PM-REG-001
- Feeds: PRJ-04; 1 downstream activities, 0 downstream deliverables

LAY-03 3D model reviews and constructability (Layout, not started)

- Needs: PIP-02 (routed model); LAY-02 (layout model); CIV-01 (structures)

- External inputs: Owner operations and maintenance reviewers
- Feeds: PRJ-04; 1 downstream activities, 0 downstream deliverables
- Note: 30/60/90 % model reviews.

ICS-08 SIL verification and SRS (I&C, not started)

- Needs: ICS-02 (SIL targets); SAF-03 (validated LOPA); VEN-01 (device failure data)
- Feeds: PRJ-04; 1 downstream activities, 0 downstream deliverables
- Note: PFDavg calculations and the safety requirements specification (IEC 61511).

CST-02 Class 3 cost estimate (Cost, not started)

- Needs: CST-01 (estimate structure); VEN-01 (budget quotes); PIP-04 (piping MTO); ELE-04 (cable quantities); CIV-01 (civil quantities)
- Feeds: PRJ-04; 1 downstream activities, 0 downstream deliverables

PRJ-04 PE review, seal and IFC issue (Project, not started)

- Needs: PRJ-03 (complete document set); SAF-03 (closed HAZOP actions); ICS-08 (SIL verification); PIP-05 (stress sign-off); ELE-05 (studies); CST-02 (Class 3 estimate); LAY-03 (model review); SAF-04 (siting study); SAF-02 (flare study)
- External inputs: Licensed Professional Engineer in responsible charge
- Feeds: nothing (end of chain); 0 downstream activities, 0 downstream deliverables
- Note: Nothing in this package is sealed; this is the gate before any use for construction.

5 Iteration loops

From	Revises	What changes
MEC-01	PRC-02	mechanical holds re-rate sizing (C-201 bed-1 diameter, heater box length)
MEC-01	LAY-01	calculated dimensions and weights update footprints and spacing
PIP-02	PRC-02	routed lengths update pump and line hydraulics
PIP-02	LAY-01	rack width and clashes move equipment
ICS-04	PRC-08	control valve sizes and reducers shown on the P&IDs
SAF-01	PRC-08	HAZOP recommendations revise the P&IDs
ELE-04	ELE-03	actual cable impedances re-run the motor-starting voltage-dip check
ICS-07	ELE-01	ICS cabinet and UPS loads added to the load list
SAF-02	PRC-03	flare back-pressure changes PSV type and size
SAF-03	PRC-08	formal HAZOP actions revise the P&IDs
SAF-04	LAY-01	siting study moves occupied buildings and equipment
VEN-01	MEC-01	certified vendor dimensions, weights and nozzles
VEN-01	LAY-01	vendor package footprints
VEN-01	ELE-01	certified motor ratings
VEN-01	ICS-05	vendor package I/O

Plan each loop as a formal update: issue the upstream activity, collect the later results, then re-issue. The vendor-data loop (VEN-01) and the HAZOP loops (SAF-01, SAF-03) usually set the FEED schedule.

6 Change impact

The activities whose change reaches the most deliverables. Freeze these first.

ID	Activity	Downstream activities	Downstream deliverables
PRC-00	Rigorous simulation and model calibration	42	104
PRJ-01	Design basis (BOD)	44	104
PRJ-02	Crude assay characterisation	43	104
PRC-01	Heat and material balance	41	103
PRC-02	Process equipment sizing	38	100
PRC-03	Relief load and PSV sizing (unit)	28	70
PRC-04	Control loops and SIF definition	24	70
LAY-01	Plot plan and equipment layout	23	59
PIP-01	Piping material classes	20	51
PRC-06	Process flow diagrams	20	51
MEC-01	Mechanical design calculations	15	38
PRC-08	P&IDs, line list, instrument index	19	31
PIP-02	Piping routing and 3D model	8	20
ELE-01	Electrical load list	7	11

7 What still needs to be done

In sequence order. "Ready" means nothing it needs is still not started. "Re-opens" counts the issued activities that must be re-checked when it completes, through its downstream links and its iteration loops.

Order	ID	Activity	Status	Ready	Re-opens
1	PRJ-02	Crude assay characterisation	Issued, provisional	yes	31 issued activities
2	PRC-00	Rigorous simulation and model calibration	Not started	yes	31 issued activities
3	PRC-01	Heat and material balance	Issued, provisional	after PRC-00	30 issued activities
4	SAF-02	Global flare load study	Not started	yes	28 issued activities
5	VEN-01	Vendor enquiries and vendor data	Not started	yes	28 issued activities
6	SAF-04	Siting study, QRA, F&G mapping	Not started	yes	28 issued activities
7	PIP-03	Pipe stress and supports (screening)	Issued, provisional	yes	1 issued activity
8	ICS-02	SIF list and SIL determination (LOPA)	Issued, provisional	yes	2 issued activities
9	ELE-05	Power system studies and protection	Not started	after VEN-01	0 issued activities
10	SAF-03	Formal HAZOP and LOPA workshop	Not started	yes	28 issued activities
11	PIP-05	Detailed pipe stress (CAESAR II)	Not started	after VEN-01	0 issued activities
12	CIV-01	Geotechnical, civil and structural design	Not started	yes	0 issued activities
13	LAY-03	3D model reviews and constructability	Not started	after CIV-01	0 issued activities
14	ICS-08	SIL verification and SRS	Not started	after SAF-03, VEN-01	0 issued activities

Order	ID	Activity	Status	Ready	Re-opens
15	CST-02	Class 3 cost estimate	Not started	after VEN-01, CIV-01	0 issued activities
16	PRJ-04	PE review, seal and IFC issue	Not started	after SAF-03, ICS-08, PIP-05, ELE-05, CST-02, LAY-03, SAF-04, SAF-02	0 issued activities

The provisional items (assay, H&MB, stress screening, SIL determination) sit at the start of long chains: replacing the shortcut process model (PRC-00, PRC-01) re-opens 41 of the 45 activities. Nothing in this package is reviewed or sealed by a licensed engineer; PRJ-04 is the gate before any use for procurement or construction.

8 Check against the code

The build was run with file access recorded (`python -m cfu.wrapup.depmap --trace, data/dataflow.json`). Each read of a data file was compared with this map:

- upstream: 62

Iterations found in the code (a module reads a file written later in the build, so it uses the previous revision):

- none

Issues:

- none: every traced read is covered by the map

9 Limits

- The map covers the activities in this FEED package and the main ones still needed. Detailed design adds more (structural steel, instrument installation details, procurement, construction planning).
- Links are finish-to-start for clarity. Real schedules overlap activities with holds and early releases.
- The code check covers data files only. Shared constants imported from `cfu/basis.py` (PRJ-01) reach every module.